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# Duskward displacement of plasmoids and reconnection in the near-Earth magnetotail

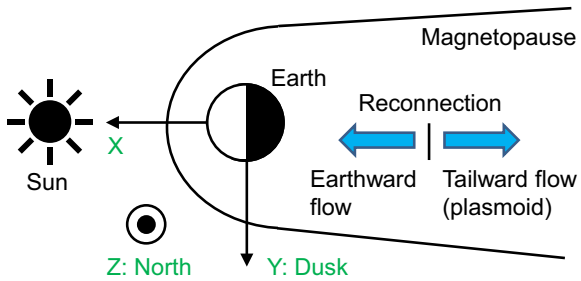
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**Abstract**

Magnetic reconnection in the near-Earth magnetotail is responsible for explosive release of energy during substorms and auroral breakups. This near-tail reconnection was previously assumed to occur around the midnight meridian, where earthward flows were typically observed. Based on observations of tailward-moving plasmoids, the Geotail spacecraft mission discovered that the reconnection location was displaced toward dusk. This dusk preference is presumably caused by the Hall electric field, as was suggested later in simulations. However, recent spacecraft observations have indicated that the reconnection was displaced toward dawn, and not dusk, in Mercury's magnetotail. In response to this controversy, our study aims to clarify the dawn–dusk location of fast plasma flows in the near-Earth magnetotail. Through a comprehensive reinterpretation and integration of previous statistical results, we found that the dusk preference is generally evident for tailward flows but is often absent for earthward flows. These results indicate that the statistical results of earthward flows are sensitive to event selection criteria. We conclude that the dawn–dusk location of earthward flow is statistically unclear at the time of substorm onset. Similarly, in the magnetotail of other planets, the dawn–dusk location of planetward flow may be sensitive to event selection criteria. Hence, reconnection may occur predominantly on the duskside in Mercury's magnetotail. This hypothesis will be tested using observations of tailward-moving plasmoids by the BepiColombo Mio spacecraft, which will begin orbital observations of Mercury in the year 2026.

**Keywords** Magnetotail, Plasmoid, Plasma flow, Reconnection, Dawn–dusk asymmetry, Auroral breakup, Substorm, Earth, Mercury

**Graphical Abstract**




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## 1 Introduction

Magnetic reconnection occurs at two locations in the magnetotail of the Earth. One reconnection site is located approximately 20 $R_E$ down the tail (Ieda et al. 1998; Miyashita et al. 1999) and is called the near-Earth neutral line (NENL). The other site is located approximately 100 $R_E$ down the tail (Slavin et al. 1985; Nishida et al. 1996) and is called the distant neutral line (DNL). In this study, we focus on the dawn–dusk location of the NENL.

The NENL model is the most popular framework for understanding substorms (Coppi et al. 1966; Atkinson 1966; Hones et al. 1973; Nishida and Nagayama 1973; Russell and McPherron 1973; Hones 1976; Baker et al. 1996; Shiokawa et al. 1998; Sergeev et al. 2012; Ieda et al. 2016). This model assumes that the NENL formation causes substorms, which are explosive energy releases and optically manifest as auroral breakups (e.g., Akasofu 1963; Ieda et al. 2018). Temporal and spatial associations between the NENL and substorm onsets have been studied to evaluate the NENL model of substorms.

Direct identification of NENL via in situ satellite observations is quite rare, particularly near the time of substorm onset, primarily due to its small size. Instead, the NENL is often identified indirectly by fast plasma flow, which is remote evidence of reconnection (Fig. 1). The NENL presumably creates a pair of fast plasma flows: an earthward flow with a northward magnetic field and a tailward flow with a southward magnetic field. According to the NENL model, this flow pattern should form at the time of substorm onset, as supported in case studies using two-satellite observations (Petrukovich et al. 1998; Petrukovich and Yahnin 2006; Slavin et al. 2002).

Prior to the Geotail satellite mission, the NENL was implicitly recognized to occur around the midnight meridian. This midnight preference of NENL was consistent with observations of earthward flows: Baumjohann et al. (1990) and Angelopoulos et al. (1994) observed that earthward flows are centered around midnight, as detailed in Sect. 3.1.

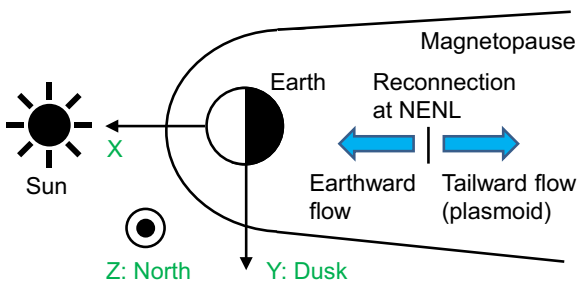



**Fig. 1** Pair of earthward and tailward flows in the Earth's magnetotail. These fast plasma flows are generated by magnetic reconnection on the near-Earth neutral line (NENL). Not to scale

In magnetohydrodynamic (MHD) simulations, the NENL was located near midnight. Hesse and Birn (1991) assumed that the plasma sheet was the thinnest near midnight under the initial conditions of their MHD simulation. The resultant earthward flow was the "fastest in the midnight region" (p. 19,418). Walker et al. (1993) reported that the reconnection first occurred near midnight in their MHD simulation. They interpreted that this midnight preference was caused by the northward component of the magnetic field being "smaller in the central part of the tail than near the magnetopause" (p. 17,241).

The Geotail spacecraft mission (Nishida 1994) discovered that the NENL is displaced toward dusk. This dusk preference was reported for the first time in two studies: Ieda et al. (1998) with tailward-moving plasmoids and Nagai et al. (1998) with fast flows. Later, this dusk preference was suggested to be caused by the Hall effect in simulations (Lin et al. 2014; Lu et al. 2018). The Hall electric field toward the north–south center is formed in the thin current sheet and transports the magnetic flux dawnward. Accordingly, the duskside plasma sheet is evacuated and further thins, leading to reconnection.

However, the MErcury Surface, Space ENvironment, GEochemistry, and Ranging (MESSENGER) spacecraft observed that the reconnection location was displaced toward dawn, not dusk, in the magnetotail of Mercury (Sun et al. 2016; Smith et al. 2017). The cause of the difference in the reconnection dawn–dusk location between Earth and Mercury remains unclear.

This report aims to clarify the dawn–dusk location of fast flows in the near-Earth magnetotail by elaborating and unifying previous observations. We found that dusk preference is a robust feature for tailward flows, but is variable among studies of earthward flows. For example, in Nagai et al. (1998), earthward flows were not displaced toward dusk but toward dawn.

The remainder of this paper is organized as follows. Section 2 reviews the discovery of dusk preference. Section 3 clarifies that the dawn–dusk location of earthward flow is variable among studies. Section 4 clarifies that no statistical studies have demonstrated the dusk preference of earthward flows within 10 min of substorm onset. Section 5 discusses why statistical results of earthward flows are sensitive to event selection criteria. Section 6 concludes that the dawn–dusk locations of earthward flows are unknown at the times of substorm onsets. Appendix A explains the aberrated geocentric solar magnetospheric (AGSM) coordinate system, which is more appropriate than the geocentric solar magnetospheric (GSM) coordinate system for studying the magnetotail. Particularly, AGSM should be adopted for studying asymmetries relative to the tail axis.

## 2 Discovery of dusk preference of reconnection in the near-Earth magnetotail

The NENL is widely recognized as displaced toward dusk (e.g., Walsh et al. 2014; Sitnov et al. 2019; Chen et al. 2019; Sun et al. 2022). This dusk preference was first reported by Ieda et al. (1998) and Nagai et al. (1998). Earthward flows were displaced toward dawn in Nagai et al. (1998).

### 2.1 Tailward-moving plasmoids by Ieda et al. (1998)

Ieda et al. (1998) found that tailward-moving plasmoids tended to be observed around $Y_{AGSM}=3$ $R_E$ in the dusk sector (Fig. 2). They concluded that "The center of plasmoids in the near tail, and possibly that of the NENL, are likely to be located around $Y_{GSM'} = 3$ $R_E$" (p. 4,463).

Detailed explanations of their analysis methods are provided below. Ieda et al. (1998) used AGSM coordinates (see Appendix A). They identified tailward-moving plasmoids in the magnetotail using Geotail observations of the magnetic field and thermal ions. The number of observed plasmoids in each $XY$ grid was normalized to the length of time of Geotail observations within 10 $R_E$ of a model neutral sheet. The resultant occurrence rates between 25 and 50 $R_E$ down the tail are shown in Fig. 2b. They fit the calculated occurrence rates using a Gaussian function. The fitted occurrence rate peaked at $Y_{AGSM}=3.2$ $R_E$ on the duskside.

### 2.2 Fast flows within 10 min of substorm onset by Nagai et al. (1998)

Nagai et al. (1998) calculated the occurrence rate of fast flows. Tailward flows tended to be observed in the dusk. Although earthward flows tended to be observed in the dawn in their figures, they did not claim this feature. Instead, they recognized that merged "tailward or earthward flows" (p. 4,424) were skewed toward dusk. They concluded that "Magnetic reconnection usually starts in the premidnight sector of the magnetotail" (p. 4,419).

Nagai et al. (1998) used AGSM coordinates. They first identified the timing of the substorm onsets using the Pi2 pulsation in the ground magnetic field. They then studied the magnetic field and thermal ions observed by Geotail within 10 min of the onsets. They classified the events into four types: (a) tailward flow events (Fig. 3a) if the 20-min interval included at least one data sample with the $X$ (sunward) component of the plasma velocity, $V_x$, $\leq -300$ km/s: (b) similarly, earthward flow events (Fig. 3b) if $V_x \geq 300$ km/s; the remaining events classified using the ion plasma beta $\beta_i$ as (c) steady-state plasma sheet events if $\beta_i \geq 0.1$; (d) lobe events if $\beta_i < 0.1$.

The occurrence rates of the tailward flows were calculated as the number of (a) tailward flow events divided by the number of [(a) + (b) + (c)] events. The occurrence rates of the (b) earthward flows were calculated similarly. These occurrence rates were calculated for each of the 5-$R_E$ bins

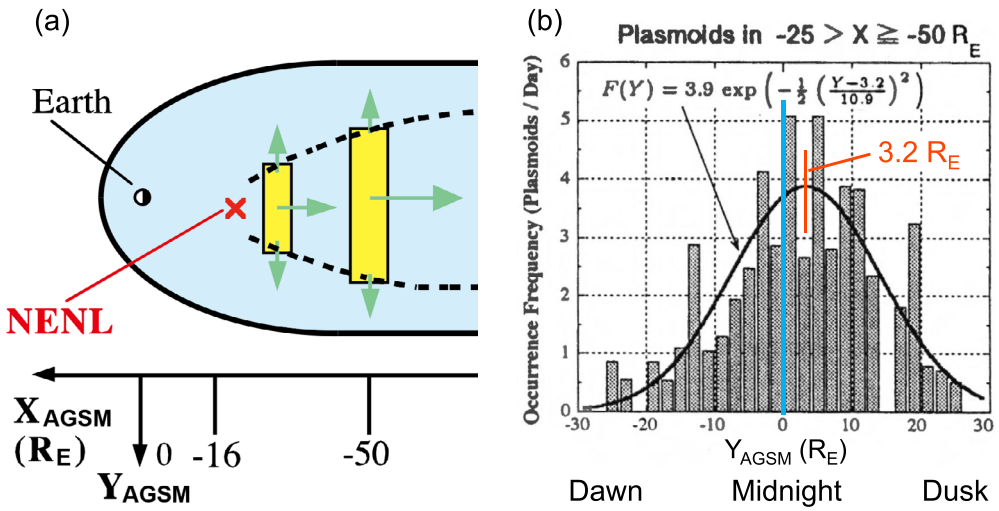



**Fig. 2** Figures adopted from Ieda et al. (1998) showing that plasmoids and reconnection are displaced toward dusk in the magnetotail. Our clarifications are added. The aberrated geocentric solar magnetospheric (AGSM) coordinate system is used. An aberration of 4° and hinging distance of 10 $R_E$ are assumed. **a** A portion of Fig. 15 from Ieda et al. (1998). The near-Earth neutral line (NENL) is inferred to be displaced toward dusk at $Y_{AGSM}=3$ $R_E$ and $X_{AGSM}=-21$ $R_E$. **b** Fig. 5d of Ieda et al. (1998). The occurrence rate of tailward-moving plasmoids, normalized by the amount of time of Geotail observations within 10 $R_E$ of $Z_{AGSM}=0$ for the time period from September 14, 1993 to March 5, 1996. Plasmoids in $-50 \leq X_{AGSM} < -25$ $R_E$ are displaced toward dusk centered at $Y_{AGSM}=3.2$ $R_E$

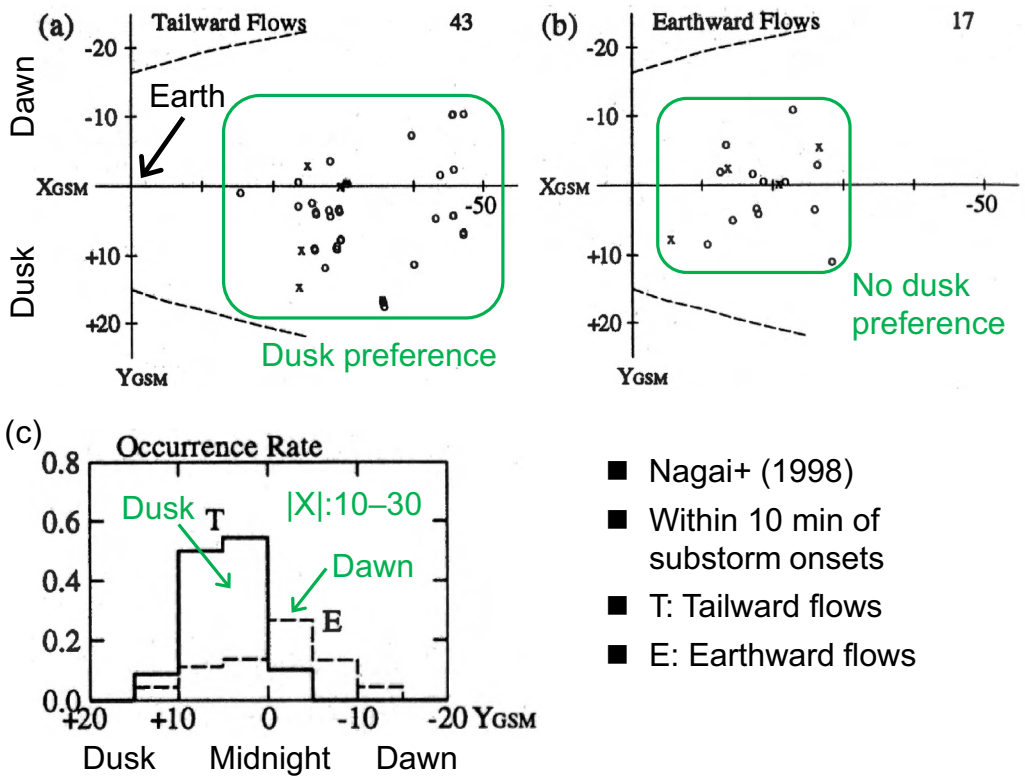



**Fig. 3** Figures 3a, b, and f from Nagai et al. (1998) showing fast plasma flow events in the magnetotail that are observed within 10 min of substorm onsets for the time period from November 1994 to July 1996. Our clarifications are added. Their figure caption states that the AGSM coordinate system is used. **a** Tailward flows and **b** earthward flows. **c** Occurrence rates of **a** and **b**. Tailward flows are displaced toward dusk and earthward flows are displaced toward dawn

in the dawn–dusk satellite locations after summation over $-30 < X_{\mathrm{AGSM}} < -10$ $R_E$ (i.e., not divided into bins in the $X$ direction). The resultant dawn–dusk distribution of the occurrence rates is shown in Fig. 3c. The occurrence rate of tailward flows has a maximum on the duskside ($0 < Y_{\mathrm{AGSM}} < 5$ $R_E$), whereas that of earthward flows has a maximum on the dawnside ($-5 < Y_{\mathrm{AGSM}} < 0$ $R_E$).

However, they did not mention the $Y$ location of earthward flows, as "Tailward flows are seen mostly in the positive $Y$ region (in the premidnight sector). Earthward flows are observed only inside 30 $R_E$" (p. 4,423). Instead, they focused on merged "tailward or earthward flows", as they recognized in the premidnight sector that "Geotail frequently observes tailward or earthward flows when the satellite is in the plasma sheet at onsets, especially beyond 20 $R_E$" (p. 4,424). That is, the earthward flow is not necessarily skewed toward dusk in their conclusions.

### 2.3 Implication of "tailward or earthward flows" in Nagai et al. (1998)

Nagai et al. (1998) concluded a dusk preference of the NENL despite the dawn preference of earthward flows in their figures. In this section, we discuss the logic behind this conclusion.

The occurrence rate was much higher for tailward flows than for earthward flows (Fig. 3c). Accordingly, "tailward or earthward flows" (Sect. 2.2) in the $-30 < X_{\mathrm{AGSM}} < -10$ $R_E$ range practically corresponded to tailward flows in their particular data set.

However, the occurrence rate is biased by the satellite $X$ locations. For example, if a satellite is more often located close to Earth (i.e., inside $X_{\mathrm{AGSM}} \sim -20$ $R_E$), the occurrence rate of earthward flows should be much higher than that of tailward flows. In such cases, "tailward or earthward flows" correspond to earthward flows.

Hence, tailward and earthward flows should be considered separately when their predominant dawn–dusk locations are distinct. The dawn preference of earthward flows in Fig. 3f of Nagai et al. (1998) (i.e., Fig. 3c of the present study) conventionally indicates that magnetic reconnection usually occurs on the dawnside, not duskside. This contradicts their conclusion.

## 3 Diverging results on the dawn–dusk location of earthward flows

Earthward flows were observed on the dawnside by Nagai et al. (1998) as described in Sect. 2.2. However, the dawn–dusk location of earthward flows is variable among studies. It can be at either dawn, midnight, or dusk, as reviewed in this section and summarized in Table 1. These discordant findings suggest that earthward flows are sensitive to the selection criteria.

### 3.1 Midnight-preferred earthward flows

Imber et al. (2011) statistically studied flux ropes using THEMIS satellite observations of magnetic fields. Tailward-moving structures were observed on the duskside (Fig. 4). In contrast, the dusk preference was not observed for earthward-moving structures inside $X = -18$ $R_E$, as they concluded that "our distribution of Earthward events becomes more symmetric within $-18$ $R_E$" (p. 11).

Angelopoulos et al. (1994) statistically studied earthward flows using AMPTE/IRM and ISEE-2 satellite observations of magnetic fields and thermal ions. They recognized that earthward flows were centered near midnight, as follows. Earthward flows were required to have a peak velocity exceeding 400 km/s. The occurrence rates of earthward flows are shown in Fig. 5. Skews may be visible to dusk ($Y_{AGSM} \sim 1$ $R_E$) in the IRM and dawn ($Y_{AGSM} \sim -4$ $R_E$) in ISEE-2. However, these skews were not mentioned by Angelopoulos et al. (1994), presumably because they were not considered statistically significant. Instead, they stated that "The spatial dependence of the earthward BBF occurrence rate on both satellites is roughly similar to the one of high speed flows in the plasma sheet described by Baumjohann et al. (1990) for the IRM data set, i.e., it tends to increase with distance from Earth and proximity to midnight." (p. 21,267). For clarification, Imber et al. (2011) likely misrecognized in their Table 1 that earthward flows

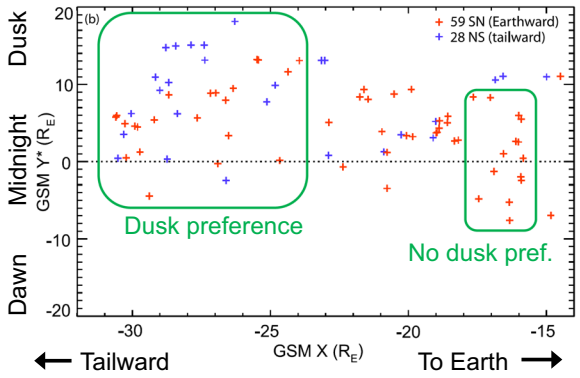



**Fig. 4** Figure 9b from Imber et al. (2011) showing locations of flux ropes and traveling compression regions in the magnetotail. Our clarifications are added. AGSM coordinates are used. "SN (Earthward)" refers to south-then-north variation in the magnetic field thus to earthward-going structures whereas "NS (tailward)" similarly refers to tailward-going structure. Earthward-going structures inside 18 $R_E$ down the tail have no dusk preference

**Table 1** Typical dawn–dusk location of plasma flows in the near-Earth magnetotail

| Reference | Dawn–dusk location | | Time from substorm onset | Spacecraft |
|---|---|---|---|---|
| | Tailward flow | Earthward flow | | |
| Baumjohann et al. (1990) | – | Midnight | – | AMTPE/IRM |
| Angelopoulos et al. (1994) | – | Midnight | – | AMPTE/IRM, ISEE-2 |
| Ieda et al. (1998) | Dusk | – | – | Geotail |
| Nagai et al. (1998) | Dusk | (Dawn) | $-10 < T < 10$ min | Geotail |
| Nagai and Machida (1998) | (Dusk) | (Dusk) | Expansion phase | Geotail |
| Nakamura et al. (2001b) | – | Midnight | – | Geotail |
| Raj et al. (2002) | – | Dusk | – | Wind |
| Ieda et al. (2008) | Dusk | – | $0 < T < 5$ min | Geotail |
| Miyashita et al. (2009) | Dusk | (Random) | (Within a few min) | Geotail |
| Imber et al. (2011)[1] | Dusk | Midnight | – | THEMIS |
| Juusola et al. (2011) | – | (Midnight) | Expansion phase | Geotail + Cluster + THEMIS |
| McPherron et al. (2011)[2] | – | Dusk | – | THEMIS |
| Li et al. (2021)[2] | – | Dusk | $-15 < T < 15$ min | THEMIS |

The dawn–dusk locations of plasma flows to remotely infer that of the near-Earth neutral line (NENL). "Location" refers to the center of statistical distribution. Parentheses indicate our interpretation of the figures. Dashes indicate that they are not applicable

[1] Imber et al. (2011) did not use the plasma data but the magnetic field data

[2] Selection criteria of earthward flows included that the flow speed inside 12 $R_E$ down the tail is above 150 km/s, and thus, selected events do not represent traditional (fast) earthward flows

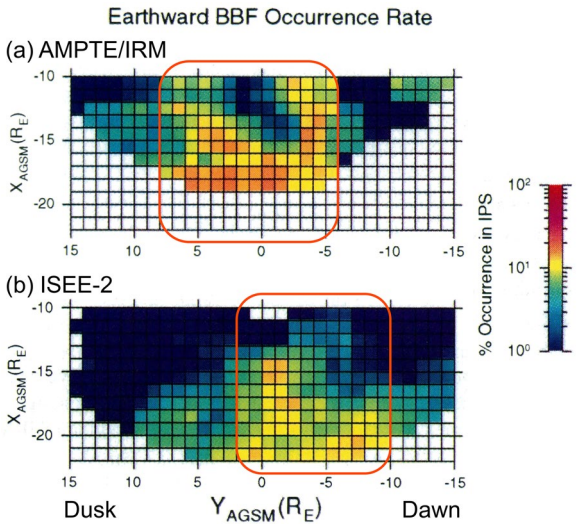



**Fig. 5** Plates 5 and 6 from Angelopoulos et al. (1994) showing locations of earthward bursty bulk flows in the magnetotail. **a** AMPTE/IRM satellite and **b** ISEE-2 satellite observations. The occurrence rate of earthward flow is normalized by the total sampling number of the inner plasma sheet. We added the red rectangles for reference where we visually see high occurrence rates. They claimed no dusk preference for the earthward flows

exhibited small skewing toward dusk in the ISEE-2 observations in Angelopoulos et al. (1994).

Nakamura et al. (2001b) statistically studied earthward flows using Geotail satellite observations of magnetic fields and thermal ions. These earthward flows did not have a dusk preference and were located near midnight, as follows. Figure 6 shows the magnetic local time (MLT) distribution of fast earthward flows. The front-side boundary of these earthward flows was assumed to be curved rather than straight. Accordingly, the azimuthal angle of earthward flow is antisymmetric between the dawn and dusk sides of the center of earthward flows. They stated that "In a statistical sense this change in the angles about the midnight region suggests that the flow is confined in the midnight region" (p. 10,798).

Juusola et al. (2011) suggested a dusk preference for earthward flows using data from Geotail, Cluster, and THEMIS satellites. They claimed that "during the substorm expansion phase, the average speed was more enhanced on the duskside than dawnside" (paragraph 39). Although they did not explicitly specify, this statement most likely refers to their Fig. 5d, where the average speed of earthward flows is shown by the colors on the *XY* plane. Although the dusk preference may slightly exist beyond 20 $R_E$ down the tail, it does not appear to exist inside 20 $R_E$, where earthward flows appear symmetric around midnight. Thus, their dusk preference is not likely associated with the NENL at the time of substorm onset (i.e., substorm expansion phase onset).

### 3.2 Dusk-preferred earthward flows

Some studies have reported a dusk preference for earthward flows. However, the dusk preference of earthward flows has not yet been statistically demonstrated within 10 min of the substorm onset.

Nagai and Machida (1998) observed that earthward flows were displaced toward dusk during the substorm expansion phase. Such earthward flows were likely observed more than 10 min after substorm onset, as detailed in Sect. 4.4.

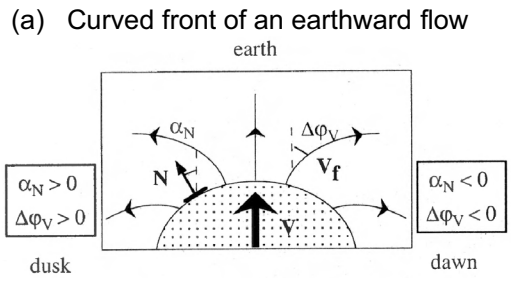



**Figure 5.** Illustration of the bubble and its front side layer adapted from *Sergeev et al.* [1996]. $\alpha_N$ is the angle between the normal component of the front side shear layer and the flow in the equatorial plane. Variable $\Delta\varphi_V$ is the difference in azimuth of the flow at the peak and flow preceding the onset of the flow burst.

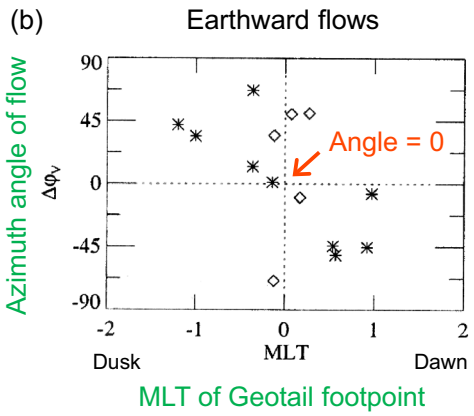



**Fig. 6** Figures from Nakamura et al. (2001b) showing earthward flows in the magnetotail. Our clarifications are added. **a** Original illustration and figure caption (Fig. 5 of their paper) explaining how to infer the dawn–dusk center of earthward flows. The sign of the azimuthal angle $\Delta\varphi_V$ of earthward flow is positive on the duskside of the dawn–dusk center and negative on the dawnside (i.e., typically negative-*Y* and positive-MLT). **b** Bottom left panel of their Fig. 6. Each symbol represents an earthward flow identified using Geotail satellite magnetic field and thermal ion data. The typical dawn–dusk center is inferred, with the zero angle condition, to be near the midnight meridian. No dusk preference was claimed for the earthward flows

Raj et al. (2002) suggested a dusk preference for fast flows. They identified flows faster than 250 km/s as measured by the Wind spacecraft inside 25 $R_E$ down the tail, regardless of geomagnetic conditions. They stated that "Essentially all bulk flows are earthward directed." (paragraph 34). They did not state the numerical value of this duskward displacement. The bottom right panel of their Fig. 12 shows the occurrence rate of fast flows against $Y_{GSM}$. The maximum occurrence rate was located at $Y_{GSM} = 14\ R_E$.

McPherron et al. (2011) concluded a dusk preference for earthward flows near $X_{GSM} = -10\ R_E$ using THEMIS D spacecraft data. Geomagnetic conditions do not appear to have been considered for this conclusion. They identified earthward flows by a peak $X$ component of the perpendicular velocity exceeding 150 km/s. This relatively slow speed was adopted as a criterion presumably because "the earthward flow speed decreases inside of 12 $R_E$" (paragraph 32). They nominally assumed that these slow "fast flows" were "produced by reconnection" (paragraph 56). They claimed that the local time distribution of their earthward flows is virtually identical to that of auroral expansion onsets reported by Frey et al. (2004).

Li et al. (2021) appear to have updated the analysis of McPherron et al. (2011) by reporting a dusk preference for earthward flows near $X_{GSM} = -10\ R_E$ using THEMIS A, D, and E spacecraft data. They identified earthward flows by a peak $XY$ component of the perpendicular velocity exceeding 150 km/s. Substorm-onset intervals were identified as within 15 min of the onset of magnetic positive bays on the ground. They observed dusk preferences during the substorm-onset and non-substorm intervals. We recognize the maximum occurrence rate of earthward flows at 22 MLT and $Y_{GSM} \sim 4\ R_E$ (i.e., $Y_{AGSM} \sim 3\ R_E$) in their Fig. 5. Their earthward flows are not traditional "fast" earthward flows, due to the slow 150 km/s criterion. Their earthward flows have likely been slowed and deflected due to interactions with the inner magnetosphere; thus, their MLT is not directly applicable to infer the MLT of NENL.

For clarification, more earthward flows were observed on the duskside than on the dawnside in Fig. 2 of Nakamura et al. (1991) and Fig. 3a of Slavin et al. (2005); however, the corresponding occurrence rate normalized by length of observation time was not shown; accordingly, no dusk preference was concluded in these studies.

## 4 Timing of earthward flows

Earthward flows are rarely observed within a few minutes of substorm onset. Instead, earthward flows are typically observed later, particularly at the time of tailward retreat of the NENL at the beginning of the substorm recovery phase. No statistical study has demonstrated the duskward displacement of earthward flow within 10 min of substorm onset.

### 4.1 Rarity of earthward flows at substorm onsets

Earthward flows are rarely identified in statistical studies around the time of substorm onset. Lui et al. (1998) showed plasma velocity vectors in the $XY$ plane observed by Geotail for 102 substorm events in their Fig. 3. Machida et al. (1999) showed similar velocity vectors for 263 substorm events in their Fig. 1. Although neither study provided occurrence rates, earthward flows were much rarer than tailward flows.

Miyashita et al. (2009) showed the average flows around the time of substorm onset in their Fig. 2. They observed that "fast tailward flows begin to grow significantly in the premidnight sector at $X < -20\ R_E$ at onset ($t = 0$)" (paragraph 17). In contrast, they did not claim a premidnight preference of earthward flows. We recognize in their Fig. 2 that the dawn–dusk location of the earthward flows appeared to change randomly with time and did not align with the longitude of the tailward flows. This random location of earthward flows is presumably due to their rarity.

### 4.2 Later earthward flows and tailward retreat of the reconnection site

Earthward flows tend to be observed much later than the onset of substorms. Nakamura et al. (2001a) summarized in their Table 1 that earthward flows were typically observed during the substorm recovery phase. Baumjohann et al. (1991) and Juusola et al. (2011) showed that earthward flows reach the maximum occurrence rate at the beginning of the substorm recovery phase, which is conventionally identified by the recovery of magnetic negative bays at auroral latitudes on the ground (e.g., Hones et al. 1973).

Earthward flows observed at the beginning of the substorm recovery phase correspond to the tailward retreat of the NENL. The NENL does not move significantly during the substorm expansion phase (Nishida and Nagayama 1973). Later, it suddenly retreats tailward at the beginning of the substorm recovery phase (Hones et al. 1973; Baumjohann et al. 1999). Ieda et al. (2016) found that this tailward retreat occurs stepwise in multiple-onset substorms. Hence, tailward retreat and associated earthward flows sometimes also occur during the middle expansion phase.

### 4.3 Direct observation of reconnection site at the beginning of the substorm recovery phase

The NENL is small and, thus, usually difficult to observe directly (i.e., locally) by satellite. The NENL is directly observed only when it moves, and thus is typically observed at the time of its tailward retreat at the beginning of the substorm recovery phase. When the NENL passes tailward over a satellite, the satellite observes a flow reversal: from a tailward flow with southward magnetic field to an earthward flow with northward magnetic field. The directly observed NENL is displaced toward dusk (Ueno et al. 1999; Nagai et al. 2015).

Ueno et al. (1999) identified 5940 passages of the NENL over the Geotail spacecraft by the flow and magnetic-field reversal. For the first time, they found that directly observed NENLs were statistically displaced toward dusk. The NENL was often observed in the region of $5 < Y_{GSM} < 15$ $R_E$ near $X_{GSM} = -28$ $R_E$. We estimate that this dusk preference should be valid under AGSM coordinates (i.e., $3 < Y_{AGSM} < 13$ $R_E$, i.e., $Y_{AGSM} \sim 8$ $R_E$), as the aberration effect is approximately 2 $R_E$.

Nagai et al. (2015) confirmed the dusk preference using a refined direct identification of 71 NENL events. In addition to the flow and magnetic-field reversal, Geotail was required to observe strong electron heating, ensuring that the satellite entered the ion–electron decoupling region. They estimated that the NENL was typically located at $3 < Y_{GSM} < 9$ $R_E$. We estimate that this corresponds to approximately $1 < Y_{AGSM} < 7$ $R_E$, i.e., $Y_{AGSM} \sim 4$ $R_E$.

The typical location of NENL at the beginning of the substorm recovery phase ($Y_{AGAM} \sim 4$ $R_E$ in Nagai et al. (2015)) is close to that at substorm onset ($Y_{AGSM} \sim 3$ $R_E$ in Ieda et al. (2008)). This closeness is in part a natural consequence as the directly observed events are a subset of tailward flow events. However, it may also be due to a coincidence caused by unknown biases in the selection of the directly observed rare events because the $Y_{AGAM}$ location of the NENL may change with time from the following points of view. One perspective is that the NENL may extend in the dawn or dusk directions in simulations (Shay et al. 2003; Nakamura et al. 2012; Liu et al. 2019). Additionally, the brightest part of auroral breakup tends to move westward (i.e., the westward traveling surge) (e.g., Akasofu 1963, 1964; Rothwell et al. 1984; Lyons et al. 2013; Ebihara and Tanaka 2015).

For clarification, we recognize that the dusk preference was not necessarily statistically significant in the following two studies. Rogers et al. (2019) identified 12 NENL events using data from the magnetospheric multiscale (MMS) spacecraft. Eleven of 12 events were on the duskside in GSM coordinates. However, based on the $Y_{GSM}$ of the NENL in their Table 2, we predict that approximately nine of 12 events were on the duskside in AGSM coordinates.

Eastwood et al. (2010) identified NENL events using data from the Cluster spacecraft and showed the locations of 18 identified events in GSM coordinates in their Fig. 4. They stated that the NENL was observed "more often on the premidnight side of the magnetotail. However, these data are a strong function of the Cluster orbit, and so one should be careful when drawing conclusions about the most typical location of the diffusion region" (paragraph 22). We recognize that by "more often" they intend to state that there were more events on the premidnight side in their Fig. 4. That is, they did not mention the corresponding occurrence rate normalized by length of observation time; accordingly, they do not presumably intend to, and indeed cannot, conclude the dusk preference of the NENL.

### 4.4 Earthward flows during substorm expansion phase by Nagai and Machida (1998)

Earthward flows are displaced toward dawn in Nagai et al. (1998). In contrast, they are displaced toward dusk in Nagai and Machida (1998) (hereinafter NM1998). In this section, we discuss that this dusk preference was caused by adding an extra interval beginning 10 min after substorm onset.

NM1998 is an article in a monograph and primarily reviews past studies, including Nagai et al. (1998). However, regarding the dawn–dusk location of the reconnection site, NM1998 conducted a new analysis in their Sect. 3.1 similar to that of Nagai et al. (1998). The selection criteria for the target interval and fast flow differ between the two studies. NM1998 did not discuss the differences in criteria and results between the two studies. Considering that NM1998 re-analyzed data and obtained the opposite result on the dawn–dusk location, they presumably recognized that the previous analysis of Nagai et al. (1998) must be revisited.

Earthward flows appear to have a dusk preference in NM1998, as shown in Fig. 7. The dusk preference is based on eight earthward flows in the dusk between 7 and 29 $R_E$ down the tail, as we indicated by the green rectangle in Fig. 7b. However, we cannot identify corresponding earthward flows in Nagai et al. (1998), as shown in Fig. 3b, except for the two events indicated by the red circles in Fig. 7b.

The criteria for the target interval were revised as follows. Nagai et al. (1998) studied the interval within 10 min of substorm onset (i.e., $-10 < T < 10$ min, where $T = 0$ refers to substorm onsets), which they identified based on the Pi2 onset in the ground magnetic field. In contrast, NM1998 studied the interval of the substorm

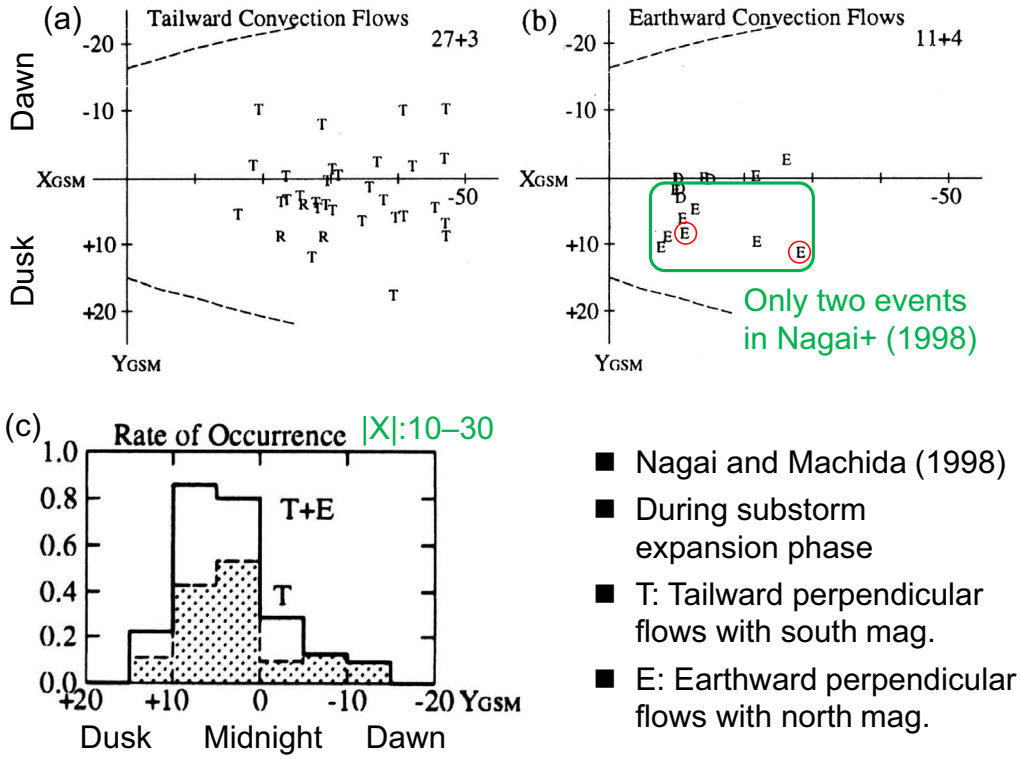



**Fig. 7** Figures 2a, b, and 3b from Nagai and Machida (1998) showing fast plasma flow events in the magnetotail that are observed during the substorm expansion phase for the time period from November 1994 to February 1996. Our clarifications are added. Their figure caption states that the AGSM coordinate system is used. **a** 27 tailward flows ("T") and 3 reversal events ("R": tailward-then-earthward flow reversal). **b** 11 earthward flows ("E") and 4 dipolarization events ("D"). **c** Occurrence rates of **a** and **b**. Earthward flows are displaced toward dusk. We added the green rectangle in **b** to indicate the dusk preference of eight "E" events between 7 and 29 $R_E$ down the tail. The red circles indicate the two events that appear shown also in Nagai el al. (1998) (i.e., Fig. 3b of the present study)

expansion phase, which they identified as the interval between the start and peak of a mid-latitude positive bay in the ground magnetic field (p. 212). Although the duration of the substorm expansion phase is not mentioned in NM1998, it is generally 20–45 min (e.g., McPherron and Chu 2018; Ieda et al. 2008; Baumjohann et al. 1991, 1999). Accordingly, NM1998 studied intervals of approximately $0 < T < 20$–$45$ min. For clarification, the interval before substorm onset was excluded from NM1998 for unknown reasons, although this interval was emphasized by Nagai et al. (1998). In summary, an interval that starts at $T = 10$ min was not included in Nagai et al. (1998), but was added in NM1998.

The criterion for earthward flow was $V_x \geq 300$ km/s in Nagai et al. (1998), as explained in Sect. 2.2. In contrast, NM1998 required a northward magnetic field and $V_{perp,x} \geq 300$ km/s, where a velocity perpendicular to the magnetic field was considered. The criteria for tailward flows were similarly updated.

Although the criteria for earthward flow are less restrictive in Nagai et al. (1998) than in NM1998, the six earthward flows are not included in Nagai et al. (1998) but are included in NM1998. Thus, the six earthward flows are likely identified during the extra interval after $T = 10$ min in NM1998.

In summary, we infer that most (i.e., six of eight) earthward flows on the duskside in Fig. 7b (i.e., Fig. 2b of NM1998) were observed more than 10 min after substorm onset. Only two earthward flows were observed on the duskside before $T = 10$ min in NM1998. Furthermore, only one earthward flow event was observed on the duskside between 13 and 28 $R_E$ down the tail in NM1998. Thus, the dusk preference of earthward flow is not demonstrated at the time of substorm onset in NM1998. In fact, Nagai et al. (1998) and NM1998 made no statement on statistical dawn–dusk locations of earthward flows in their text.

## 5 Discussion

Sections 5.1–5.3 discuss why the dawn–dusk location of earthward flows is sensitive to event selection criteria. Sections 5.4–5.5 confirm that plasma injection, magnetic dipolarization, and auroral breakup are displaced toward dusk. These phenomena can be located duskside of the NENL in terms of the substorm current wedge and by the possible deflection of earthward flows near $X_{AGSM} = -10$ $R_E$. Section 5.6 offers possible explanations for the dawn–dusk location of the reconnection site that differs between Earth and Mercury.

### 5.1 Paired earthward and tailward flows not yet demonstrated

As mentioned in Sect. 1, the NENL model of substorms requires a pair of earthward and tailward flows at the time of substorm onset. In this model, the dawn–dusk location of the two flows is implicitly assumed to be the same (Fig. 1). Although this assumption is presumably correct, this feature has not yet been statistically demonstrated for the time of substorm onset, possibly due to observational biases, as follows.

In Sect. 2, tailward flows were discovered to be displaced toward dusk. This preference has been robustly observed (e.g., Ieda et al. 1998, 2008; Nagai et al. 1998; Machida et al. 1999; Miyashita et al. 2009; Imber et al. 2011). Accordingly, the NENL forms on the duskside.

In contrast, the dawn–dusk location of earthward flows varies among studies. It can be either dawn, midnight, or dusk, as described in Sect. 3. In particular, no statistical study has demonstrated the duskward displacement of earthward flows within 10 min of substorm onset, as discussed in Sects. 3.2 and 4. These discordant findings indicate that the statistical results of earthward flows are sensitive to event selection criteria.

This sensitivity is presumably due to observational biases in earthward flows. One potential bias is caused by the fact that earthward flows do not necessarily originate from the NENL, as detailed in Sect. 5.2. Another potential bias is caused by the variability in the thickness in $Z_{AGSM}$ of earthward flows (Sect. 5.3). In addition, earthward flows are narrow and typically 2 $R_E$ in $Y_{AGSM}$ in multi-spacecraft observations (Sergeev et al. 1996; Nakamura et al. 2004), causing them to be missed, depending on selection criteria. They are much narrower than tailward flows, which are typically 10 $R_E$ in $Y_{AGSM}$ or 2 h in MLT at $X_{AGSM} \sim -28$ $R_E$ (Ieda et al. 2008).

Nagai et al. (1998) found that earthward flows were displaced toward dawn. Although this was not reflected in NM1998 and is not widely appreciated in later studies, it can imply valuable information as a consequence of the adopted selection criteria. We propose one possible interpretation, involving a potential bias caused by a dawn–dusk asymmetry of the plasma sheet thickness. The plasma sheet is presumably thinner on the duskside than on the dawnside inside 20 $R_E$ down the tail before substorm onset, as can be inferred from observations of the northward magnetic field (Miyashita et al. 2009, paragraph 20) and current density (Petrukovich et al. 2009, paragraph 21). Accordingly, onset-associated earthward flows can be observed more readily on the dawnside or midnight even if they exist more often on the duskside.

For clarification, dawn–dusk asymmetries of the thin current sheet were reported by Rong et al. (2011), Artemyev et al. (2011), and Rogers et al. (2023). These studies do not mention the timing of the thin current sheet observation relative to substorm onset. The current sheet generally becomes very thin after the onset of reconnection and substorm, due to the evacuation of the plasma. Thus, the asymmetries were not necessarily observed before substorm onsets.

### 5.2 Tailward flows as strong evidence of near-tail reconnection

Tailward flows provide strong evidence of reconnection. This is because tailward flows are typically accompanied by a southward magnetic field, which requires reconnection. In contrast, earthward flows provide weaker evidence of reconnection as they are typically accompanied by a northward magnetic field, which originates from the Earth's main field and thus does not require reconnection. For example, Lui (2009) reported that some earthward flows are caused by the near-Earth current disruption without tail reconnection.

Furthermore, tailward flows inside ~30 $R_E$ should originate from the NENL and not the DNL. The tailward flows typically occur within a few minutes of substorm onset (Moldwin and Hughes 1993; Nagai et al. 1994; Ieda et al. 2008). In contrast, earthward flows inside ~ 30 $R_E$ do not necessarily originate from the NENL but can originate from the DNL, which is symmetric around $Y_{AGSM} = 0$ (Slavin et al. 1985). Earthward flows are rarely observed at the time of substorm onset (Sect. 4). Consequently, tailward flows are better associated with global auroral activity compared to earthward flows are (Shue et al. 2003, 2008; Ieda et al. 2003, 2005) although both earthward flows (Fairfield et al. 1999; Nakamura et al. 2001a) and tailward-moving plasmoids (Ieda et al. 2001) are consistently associated with auroral activity near the satellite footpoint.

The tailward flow from the NENL includes a magnetic loop called a plasmoid, which is observed by the spacecraft as a north-then-south variation in the magnetic

field. The northward magnetic field in the leading half of plasmoids is evidence that plasmoids originate from the NENL, not the DNL. Accordingly, plasmoids provide the strongest evidence of the NENL near substorm onset. However, any tailward flows well inside 100 $R_E$ down the tail are unlikely to originate from the DNL by definition. Such tailward flows with a southward magnetic field are practically equivalent to tailward-moving plasmoids in terms of the strong NENL evidence, in particular when tailward flows are relatively fast.

The dusk preference of the NENL was discovered by Geotail, which enabled the first statistical studies of tailward flows. Geotail was the first satellite that intensively observed the equatorial magnetotail beyond 20 $R_E$ with appropriate thermal ion (Mukai et al. 1994) and magnetic field (Kokubun et al. 1994) instruments. The dusk preference of the NENL was not recognized by previous satellites like AMPTE/IRM and ISEE-2, presumably because tailward flows were rarely observed due to relatively low apogees inside approximately 20 $R_E$. This indicates that the dawn–dusk location of NENL is difficult to conclude from statistical observations of earthward flows.

### 5.3 North–south thickness of flows

The dusk preference of tailward flows is robustly observed, presumably because tailward flows are thick in the $Z_{AGSM}$ direction. The $Y_{AGSM}$ location of earthward flows varies among studies, presumably because earthward flows can be thin or thick. We discuss the thickness of earthward and tailward flows in this section.

Statistical results of Asano et al. (2004) suggest that earthward flows are thinner than tailward flows at the time of the substorm onset. They statistically studied the spatial distribution of the thin current sheet in the substorm growth phase using Geotail data. They found that "the region where current sheet thinning occurs is located only on the Earthward side of the initial X line position. In other words, the initial X line forms near the tailward edge of the evolved thin current sheet." (paragraph 14).

Earthward flows tend to be more often observed at $X_{AGSM} \sim -10$ $R_E$ than at $X_{AGSM} \sim -16$ $R_E$ around the times of substorm onset. A pair of earthward and tailward flows were observed in case studies using multi-satellite observations (Petrukovich et al. 1998; Slavin et al. 2002; Angelopoulos et al. 2008). These earthward flows were observed at approximately $-12 < X_{AGSM} < -9$ $R_E$. A satellite at $X_{AGSM} = -17$ $R_E$ did not observe the corresponding earthward flow in the event reported by Angelopoulos et al. (2008).

These case studies suggest that earthward flows are thin at $X_{AGSM} \sim -16$ $R_E$ and simultaneously thick at $X_{AGSM} \sim -10$ $R_E$ around the times of substorm onset. Note, as a representative example, Asano et al. (2003) conducted a case study of the thin current sheet in a substorm observed by Geotail at $X_{GSM} = -15$ $R_E$. Earthward flows are opposed by the inner magnetosphere and slow down near the dipole-tail boundary around $X_{AGSM} = -10$ $R_E$. Earthward flows presumably thicken due to this interaction. Earthward flows are also observed at the tailward retreat (Sect. 4.2) at the beginning of the substorm recovery phase, which is accompanied by the thickening of the plasma sheet (Hones et al. 1973).

### 5.4 Dusk preference of injection and dipolarization

Plasma injection and magnetic dipolarization are reportedly observed to be displaced toward dusk between approximately 7–12 $R_E$ down the tail.

Arnoldy and Moore (1983) observed a dusk preference for plasma injection and magnetic dipolarization in a geosynchronous orbit (6.6 $R_E$). Using measurements from two GOES satellites, they studied the synchronous orbital manifestation of substorms, that is, plasma injection and dipolar magnetic field reconfiguration. They found that "the earliest onset sector at synchronous orbit is about 3 hour wide, skewed toward the evening side of midnight" (p. 6213). Nagai et al. (1983) conducted a similar study and reached a similar conclusion that states "Substorm activity initially starts in a longitudinally limited sector that is situated in the premidnight region" (p. 6,994).

Miyashita et al. (2009) statistically studied magnetic dipolarization by calculating the time evolution of the average magnetic field. The dipolarization began around the time of the substorm onset at $X_{GSM} \sim -7$ to $-10$ $R_E$ and $Y_{GSM} \sim 4$ $R_E$, which corresponds to $Y_{AGSM} \sim 3$ $R_E$ when the aberration of $\sim 0.7$ $R_E$ is considered.

Liu et al. (2013) statistically studied magnetic dipolarization, which they called "dipolarizing flux bundles." They identified dipolarization events by rapid increases in the northward magnetic field. They did not study the timing of dipolarization relative to substorm onset. The occurrence rate of events was normalized by the THEMIS satellite orbit time in each bin. They showed that the resultant location of dipolarization was displaced toward dusk in GSM coordinates. Although they did not state the magnitude of this displacement, the occurrence rate appears the highest in the $Y_{GSM} = 2$–$6$ $R_E$ bin at $X_{GSM} = -10$ $R_E$ in their Fig. 2c, thus at $Y_{AGSM} \sim 1$–$5 \sim 3$ $R_E$.

### 5.5 Dusk preference of auroral breakup

Using global images onboard the Dynamics Explorer 1 satellite, Craven and Frank (1991) found that auroral

breakups tend to occur on the duskside. They stated that "These observations confirm the well-known but not well-documented fact that auroral substorm onsets are observed most frequently late in the premidnight sector" (p. 284). They recognized that this MLT around 22.8 MLT is nearly identical to the 23 MLT of the plasma injection in Fig. 7 of Arnoldy and Moore (1983).

Corresponding fast flows and the NENL are suggested to be displaced to dawn from breakup MLTs. Nakamura et al. (2001b) found that earthward flows are displaced to dawn by 0.4 h in MLT from the center of auroral expansion based on their statistical study. Ieda et al. (2008) suggested that the *Y*-center ($Y_{AGSM}=3$ $R_E$) of tailward flows at $X_{AGSM}=-28$ $R_E$ was displaced toward dawn by 0.3 h in MLT from auroral breakups. This displacement was not statistically significant because it was much shorter than the typical width of tailward flows (2 h in MLT). Nagai and Shinohara (2021) found that the NENL is located duskward of the center of the substorm current wedge; however, we recognize that the NENL can still be dawnward of the duskside edge of the current wedge.

One interpretation of the displacement between the NENL and breakup is that auroral brightenings occur on the duskside edge of the current wedge (e.g., Amm and Fujii 2008; Kepko et al. 2015). We propose an alternative interpretation wherein the MLT of earthward flows decreases as they progress in the dusk. This decrease in MLT persists even if the streamline follows the $X_{AGSM}$ direction. The MLT further decreases if earthward flows curve duskward near $X_{AGSM}=-10$ $R_E$ due to the interaction with the inner magnetosphere. If the NENL is located on the duskside, the resultant earthward flows on the duskside are presumably deflected further toward dusk near $X_{AGSM}=-10$ $R_E$. The MLT of earthward flows inside $X_{AGSM}\sim-10$ $R_E$ is presumably close to the MLT of auroral breakups (McPherron et al. 2011). In summary, the NENL can be dawnward of the auroral breakup, injection, and dipolarization.

### 5.6 Mercury and Earth

As explained in Sect. 1, the reconnection site is recognized as being displaced toward dusk at Earth and toward dawn at Mercury. Here, we review two possibilities responsible for the dawn preference of Mercury and propose an alternative hypothesis.

One possible cause for the dawn preference is a local asymmetry within the thin current sheet (TCS), as proposed by Liu et al. (2019). They conducted three-dimensional particle-in-cell local simulations of the TCS, the length of which is assumed to be fixed in the dawn–dusk direction. The magnetic flux is transported dawnward by the Hall effect, as was explained in Sect. 1. They found that reconnection is active only in the dawnside portion of the TCS. In contrast, reconnection is suppressed in the duskside portion because an excessive removal of the north–south component of the magnetic field suppresses the generation of outward flows in the Sun–tail direction, ultimately preventing the further thinning of the current sheet to the electron scale.

This local dawn preference inside the TCS is negligible in the Earth's magnetotail as the TCS is considerably shorter than the whole current sheet in the dawn–dusk direction. That is, both the dawnside and duskside portions of the TCS tend to be located within the dusk sector. In contrast, the TCS is comparable in length to the whole current sheet in the Mercury's magnetotail due to the short dawn–dusk length of the Mercury's magnetotail. That is, there is no room for the TCS to skew significantly toward dusk; hence, the dawnside portion of the TCS tends to be located in the dawn and the duskside portion in the dusk. For clarification, the simulations explained in Sect. 1 (Lin et al. 2014; Lu et al. 2018) compare the thin and the thick current sheet, not the two portions inside the TCS.

Another possible cause is a strong southward interplanetary magnetic field (IMF), as proposed by Chen et al. (2019). They combined global MHD and local particle-in-cell simulations and found the dawn preference to be more evident for a stronger southward IMF. The background current sheet is thicker in the dawn than in the dusk. The strong IMF compresses the thick current sheet in the dawn so that the background condition is improved to allow for reconnection. This model requires the local dawn preference proposed by Liu et al. (2019).

The Chen et al. (2019) model is partly consistent with the following observations. Liou et al. (2001) found a weak correlation between the southward IMF and the MLT of auroral breakups on Earth. Ieda et al. (2008) found a correlation between the MLT of auroral breakups on Earth and that of tailward flows (i.e., NENL). Thus, the MLT of the NENL is expected to correlate with the southward IMF. This is consistent with the results of Nagai and Shinohara (2022), who showed that the NENL in the postmidnight sector occurred under strong solar wind energy input conditions.

We propose an alternative hypothesis that reconnection tends to occur in the dusk at Mercury as well as at Earth. As discussed in Sect. 5.1, the dusk preference of tailward flows is robustly observed in the Earth's magnetotail. In contrast, the dusk preference of earthward flows is unclear or absent in many previous statistical studies. Thus, the NENL is not necessarily located on the meridian where earthward flows are observed. Similarly, in the magnetotail of other planets the reconnection site may not be located on the meridian where planetward flows

are observed. The dawn preference of the reconnection site in the Mercury's magnetotail has been inferred primarily using magnetic field data (Sun et al. 2016; Smith et al. 2017). This observation is subject to further confirmation based on thermal plasma data that will be acquired in the future. Tailward-moving plasmoids may be displaced toward dusk in the magnetotail of Mercury.

## 6 Conclusions

The NENL has been widely recognized as displaced toward dusk (e.g., Walsh et al. 2014; Sitnov et al. 2019; Chen et al. 2019; Sun et al. 2022). The present paper reviewed and integrated previous observations of the dawn–dusk location of fast plasma flows in the near-Earth magnetotail. We found that the dusk preference has been robustly reported for tailward flows but has been often absent for earthward flows.

These discordant observations indicate that the statistical results of earthward flows are sensitive to event selection criteria, presumably due to observational biases. One potential bias is caused by the fact that earthward flows do not necessarily originate from the NENL, but also from the DNL. Another potential bias is caused by the variability in the thickness of earthward flows.

Earthward flows were rarely observed at the time of substorm onset and generally observed later. No statistical study has demonstrated that earthward flows are displaced toward dusk within 10 min of the substorm onset. We conclude that the dawn–dusk locations of earthward flows are statistically unclear at the time of substorm onset.

The results of this study suggest that reconnection in the near-planet magnetotail does not necessarily form on the meridian where planetward flows are observed. Hence, the tailward-moving plasmoid and the reconnection site may have been predominantly displaced toward dusk in the magnetotail of Mercury. This hypothesis will be tested using thermal plasma measurements, in addition to magnetic field measurements, from the BepiColombo Mio spacecraft.

## Appendix A: AGSM coordinate system for the magnetotail

The geocentric solar magnetospheric (GSM) coordinate system is often used to study the magnetosphere. However, the GSM system does not consider the orbital motion of the Earth about the Sun. The actual tail axis is typically 4° aberrated from the Sun–Earth axis by the windsock effect. Such a typical tail axis is adopted as the X axis in the aberrated GSM (AGSM) coordinate system (Fairfield 1979; Yamamoto et al. 1994; Angelopoulos et al. 1994; Ieda et al. 1998). The AGSM system is more appropriate for studying the magnetotail than the GSM system and should be adopted for studying asymmetries relative to the tail axis.

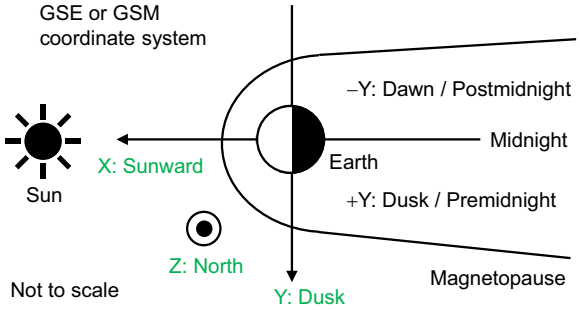



**Fig. 8** Illustration of two similar coordinate systems: the geocentric solar ecliptic (GSE) system and the geocentric solar magnetospheric (GSM) system. The two systems have the same origin (the center of Earth) and *X* axis (sunward). The *Y* (duskward) and *Z* axes (northward) slightly differ between the two systems and are defined using the ecliptic plane (i.e., orbital plane of Earth around the Sun) for GSE and the geomagnetic dipole for GSM. Further explanation can be found in Appendices A.3.3.4 and A.3.3.6 of Russell et al. (2016)

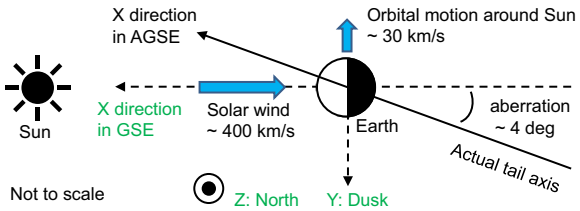



**Fig. 9** Illustration of the aberration of the magnetotail in GSE. This aberration is caused by the orbital motion of the Earth around the Sun. See definitions of the GSE and GSM coordinate systems in Fig. 8 caption. GSE is converted to the aberrated GSE (AGSE) system by a rotation in the *XY* plane. AGSE is further converted to the aberrated GSM (AGSM) system by a rotation in the *YZ* plane similar to the conversion from GSE to GSM. AGSM is more appropriate than GSM for studying the magnetotail

Figure 8 shows the GSM coordinate system. This illustration also applies to the geocentric solar ecliptic (GSE) coordinate system. The magnetosphere is bordered by a line labeled "Magnetopause". Y = 0 corresponds to midnight in the tail. The positive Y region corresponds to dusk, which is also referred to as premidnight.

Figure 9 illustrates the aberration of the actual tail axis in the GSE system. On average, the solar wind flows in the negative $X_{GSE}$ direction. The Earth moves in the negative $Y_{GSE}$ direction in an orbital motion around the Sun. Due to this orbital motion, the solar wind does not flow exactly in the negative $X_{GSE}$ direction but in an oblique direction in frames moving with the Earth. The windsock effect aligns the tail axis direction to the oblique solar wind direction. That is, the actual tail axis is not exactly aligned with the $X_{GSE}$ direction, but is aberrated.

**Table 2** The aberration of the tail axis from the Sun–Earth axis

| Solar wind speed | Aberration angle | Displacement in the *Y* direction d*Y* ($R_E$) | | | | |
|---|---|---|---|---|---|---|
| Vsw (km/s) | φ (deg) | X = −10 | X = −20 | X = −30 | X = −100 | X = −200 |
| 300 | 5.67 | 0.99 | 1.99 | 2.98 | 9.93 | 19.85 |
| 378.4 | 4.5 | 0.79 | 1.57 | 2.36 | 7.87 | 15.74 |
| 400 | 4.26 | 0.74 | 1.49 | 2.23 | 7.45 | 14.89 |
| 425.9 | 4 | 0.70 | 1.40 | 2.10 | 6.99 | 13.99 |
| 500 | 3.41 | 0.60 | 1.19 | 1.79 | 5.96 | 11.91 |
| 800 | 2.13 | 0.37 | 0.74 | 1.12 | 3.72 | 7.45 |

The geocentric solar ecliptic (GSE) coordinate system is used. The aberration is caused by the orbital motion of the Earth about Sun ($V_E$ = 29.78 km/s). The aberration angle φ (rad) of the tail axis in the second column is calculated as a function of the speed of solar wind $V_{SW}$ using $V_E = V_{SW}$ tan φ. The resultant displacement of the tail axis d$Y$ is further calculated as a function of the downtail distance $|-X|$ using d$Y = |-X|$ tan φ. The unit is nominally in $R_E$ (Earth radius, 6371.2 km) but can be arbitrary

The aberration depends on the solar wind velocity, as summarized in Table 2. The average orbital speed is generally 29.78 km/s. An aberration of 4° is often used (e.g., Yamamoto et al. 1994; Ieda et al. 1998) and corresponds to a solar wind speed of 426 km/s and a 2.1 $R_E$ displacement of the tail axis toward dusk at $X = -30$ $R_E$ at midnight.

The aberration should be considered in the GSE, not GSM, by definition, although we are not aware of explicit explanations in literature. Otherwise, the resultant $Z_{AGSM}$ location may have an artificial dawn–dusk asymmetry, which may have significant impacts because the plasma sheet is thin in the $Z$ direction. Satellite locations and measured vector data should be prepared in GSE, then converted into aberrated GSE (AGSE), and finally into AGSM.

The $Z_{AGSM}$ location is often further corrected to represent the relative distance from the actual equatorial plane (i.e., neutral sheet) of the magnetotail. To emphasize this correction, the coordinate system was sometimes labeled GSM′ in the past. The simplest neutral sheet model assumes a hinging typically at 10 $R_E$ geocentric distance, as adopted by Yamamoto et al. (1994) and Ieda et al. (1998): The tail axis is assumed to be perpendicular to the geomagnetic dipole within a geocentric distance of 10 $R_E$ and parallel to the $X_{AGSM}$ axis at greater distances. A more refined neutral sheet model is the Tsyganenko and Fairfield (2004) model, as adopted by Ieda et al. (2016).

**Abbreviations**

| | |
|---|---|
| AGSM | Aberrated geocentric solar magnetospheric |
| AMPTE/IRM | Active magnetospheric particle tracer explorers ion release module |
| DNL | Distant neutral line |
| IMF | Interplanetary magnetic field |
| ISEE-2 | International Sun–Earth explorer 2 |
| GOES | Geostationary operational environmental satellite |
| GSE | Geocentric solar ecliptic |
| GSM | Geocentric solar magnetospheric |
| MESSENGER | Mercury surface, space environment, geochemistry, and ranging |
| MHD | Magnetohydrodynamic |
| MLT | Magnetic local time in hours |
| NENL | Near-Earth neutral line |
| TCS | Thin current sheet |
| THEMIS | Time history of events and macroscale interactions during substorms |

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**Author contributions**
AI planned this study and drafted the manuscript. YM contributed to data interpretation and discussion. All authors read and approved the final manuscript.

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**Availability of data and materials**
No data were used in this study.

## Declarations

**Competing interests**
The author declares that he has no competing interests.

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